

VoiceGuard

AI Voice Detection — Evaluation Report

Model: voiceguard_cnn_finetuned.pt

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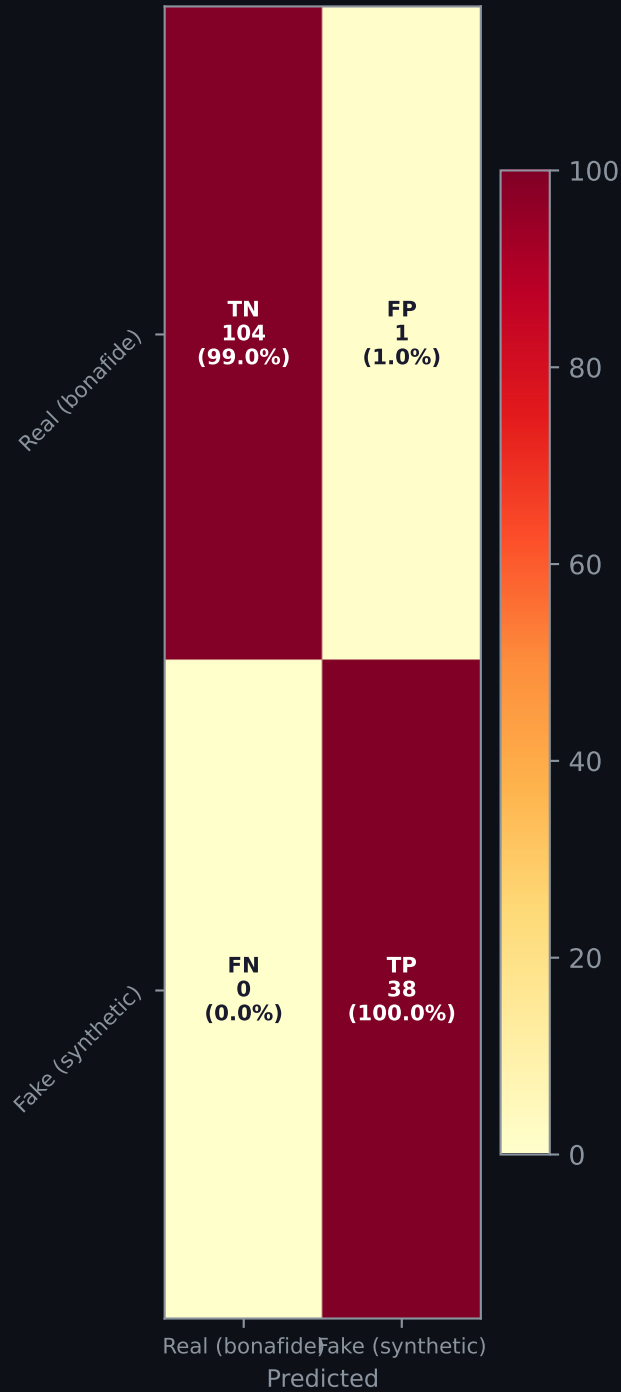
99.3%	98.7%	97.4%	100.0%	0.95%
Accuracy	F1-Score	Precision	Recall	EER

Based on personal mic validation set

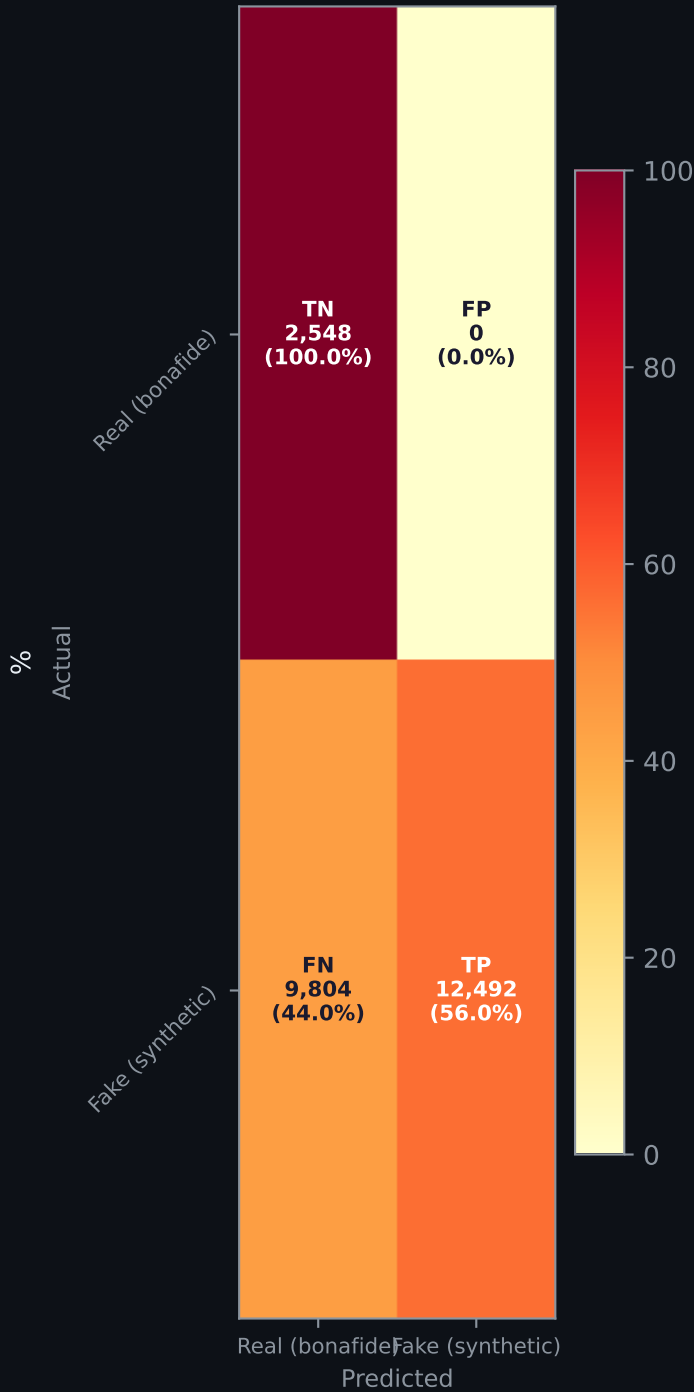
Dataset	Clips	Accuracy	F1	Precision	Recall	EER
Personal Mic Validation	143	99.30%	98.70%	97.44%	100.00%	0.95%
ASVspoof 2019 LA Dev	24,844	60.54%	71.82%	100.00%	56.03%	0.98%
ASVspoof 2021 LA Eval (max 3000/class)	6,000	61.88%	39.86%	94.40%	25.27%	30.83%

Confusion Matrices

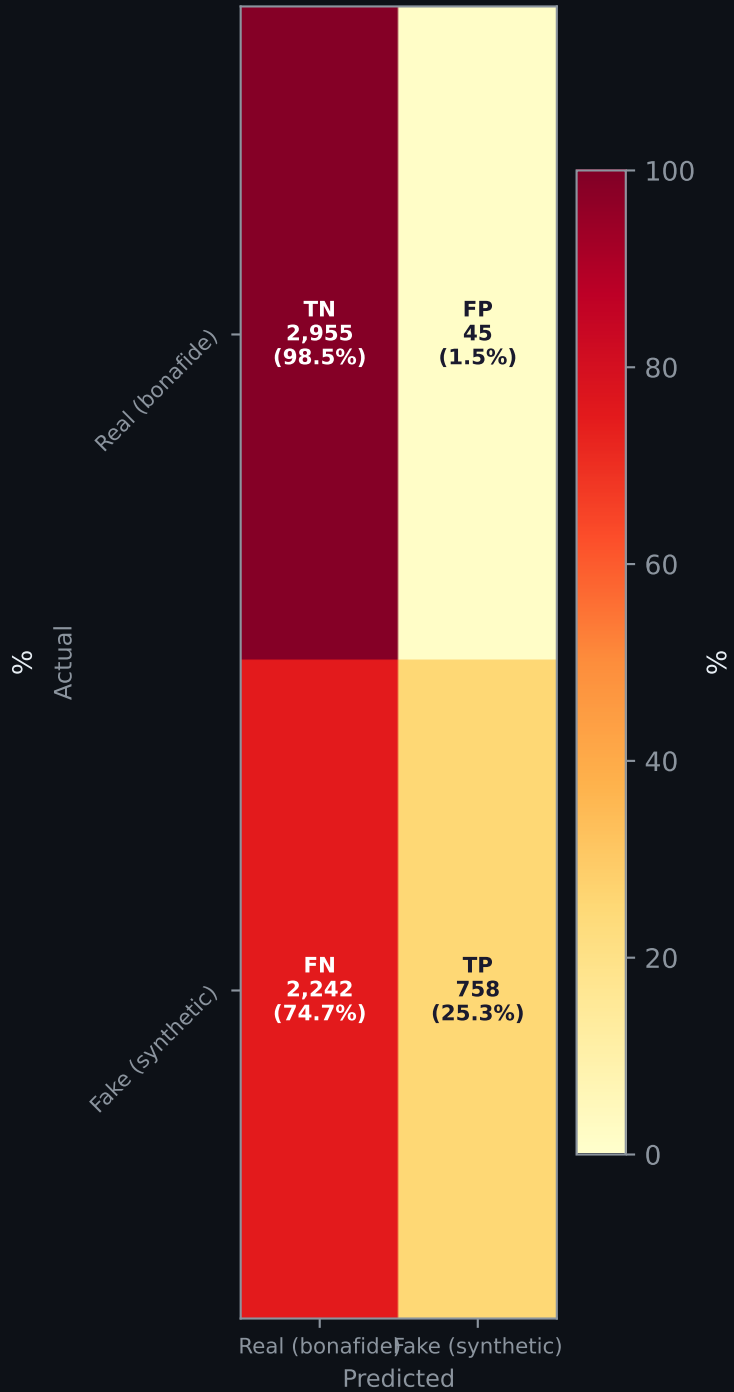
Personal Mic Validation



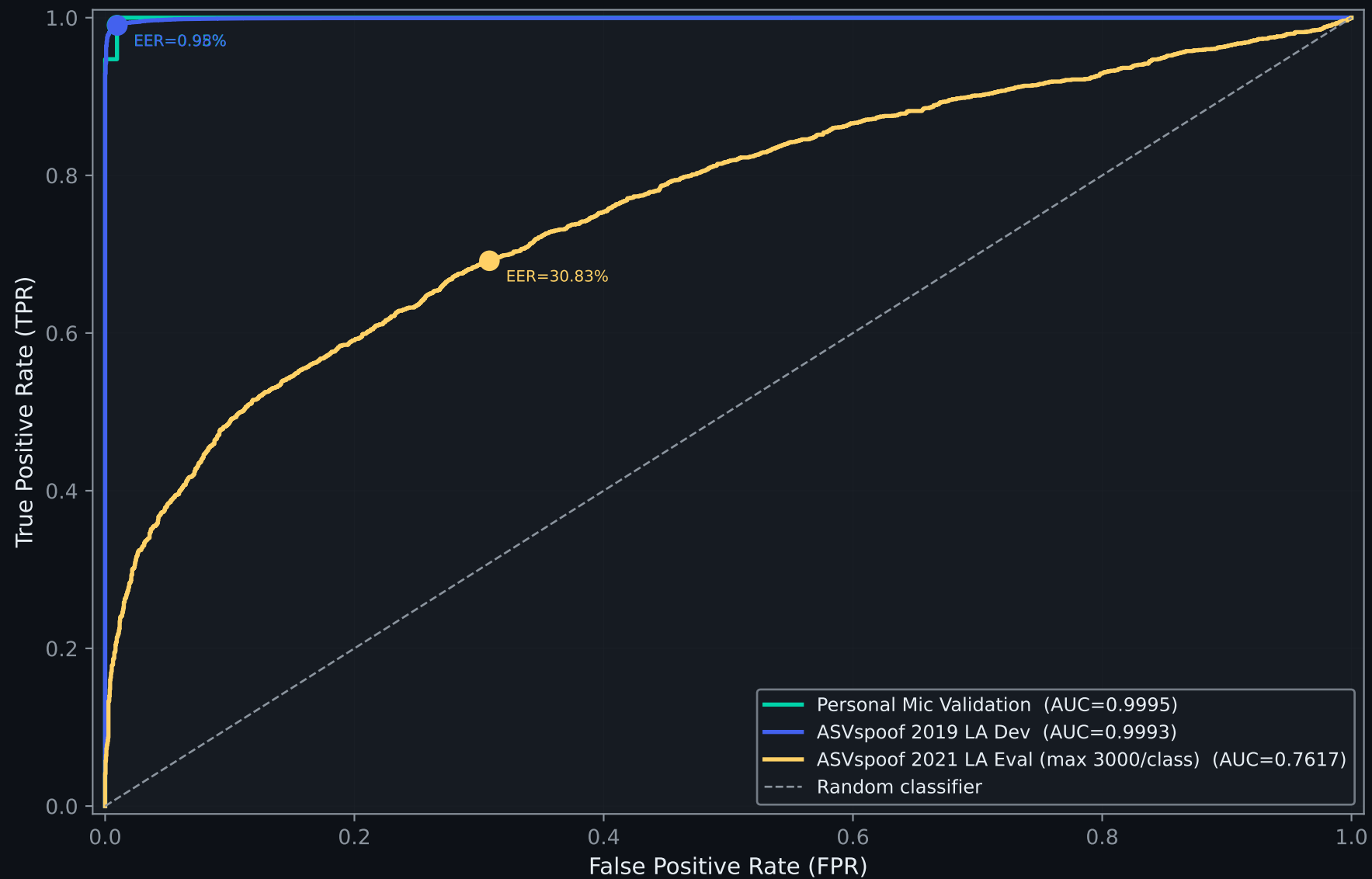
ASVspoof 2019 LA Dev



ASVspoof 2021 LA Eval (max 3000/class)

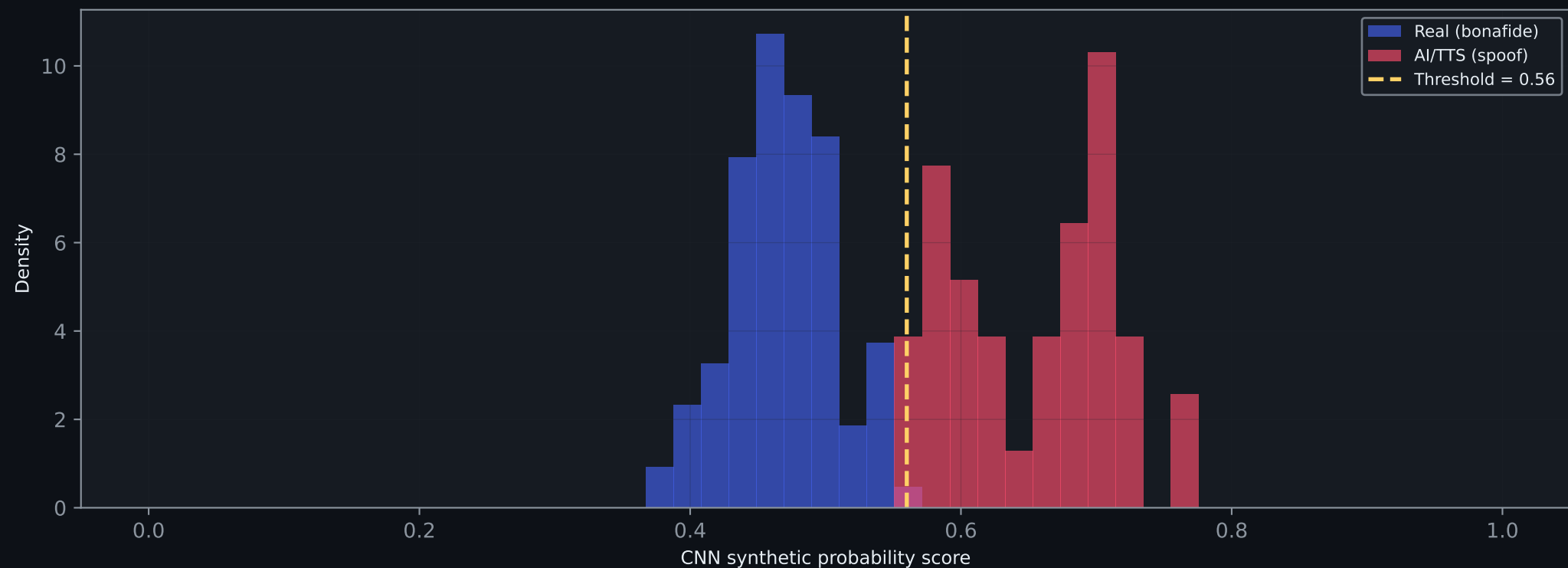


ROC Curves — All Datasets

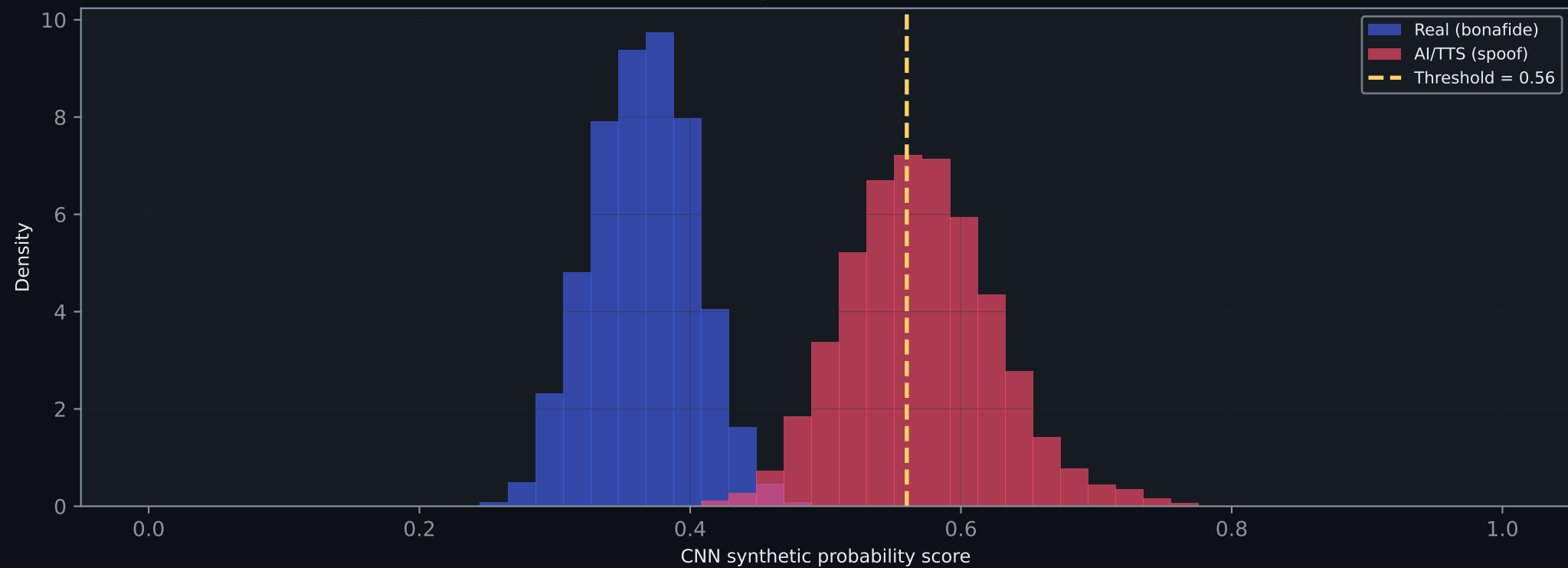


CNN Score Distributions (threshold = 0.56)

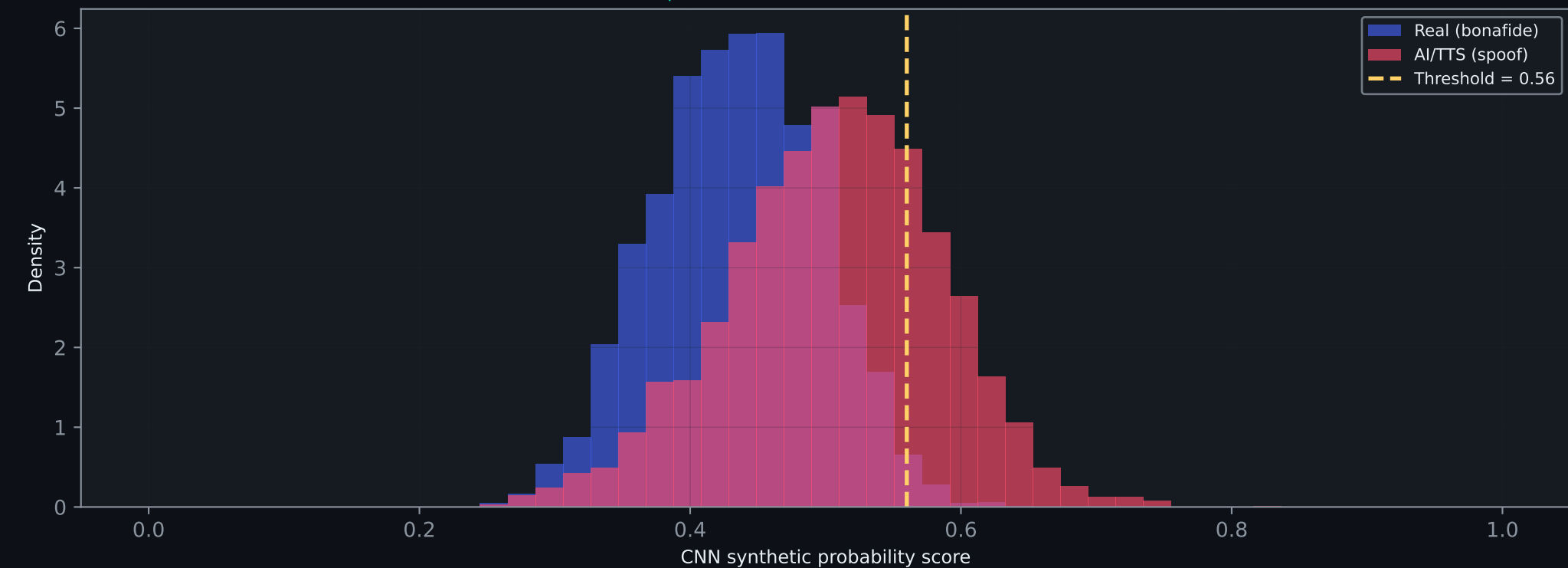
Personal Mic Validation



ASVspoof 2019 LA Dev



ASVspoof 2021 LA Eval (max 3000/class)



Detailed Metrics by Dataset

Dataset	Clips	Real	Fake	TP	TN	FP	FN	Accuracy	Precision	Recall	F1-Score	EER	AUC-ROC	FPR	FNR
Personal Mic Validation	143	105	38	38	104	1	0	99.30%	97.44%	100.00%	98.70%	0.95%	0.9995	0.95%	0.00%
ASVspoof 2019 LA Dev	24,844	2,548	22,296	12,492	2,548	0	9,804	60.54%	100.00%	56.03%	71.82%	0.98%	0.9993	0.00%	43.97%
ASVspoof 2021 LA Eval (max 30000 class)	6,000	3,000	3,000	758	2,955	45	2,242	61.88%	94.40%	25.27%	39.86%	30.83%	0.7617	1.50%	74.73%

FPR = Real voice wrongly flagged as AI | FNR = AI voice that slipped through | EER = Equal Error Rate (lower is better) | AUC-ROC closer to 1.0 is better

Model Architecture & Training Summary

VoiceGuardCNN Architecture

- Input : Log-mel spectrogram (B, 1, 128, T) — 128 mel bins, 10 ms hop, 16 kHz
- Block 1: Conv2d(1→32, 3×3) → BatchNorm → ReLU → MaxPool2d(2×2)
- Block 2: Conv2d(32→64, 3×3) → BatchNorm → ReLU → MaxPool2d(2×2)
- Block 3: Conv2d(64→128, 3×3) → BatchNorm → ReLU → MaxPool2d(2×2)
- Global Average Pool → Flatten → FC(128→1) → Sigmoid
- Output: P(synthetic) ∈ [0, 1] | Threshold: 0.56

Training Datasets

- Phase 1 — ASVspoof 2019 LA: 2,580 bonafide + 22,800 spoof (train split)
- Phase 2 — ASVspoof 2021 LA: 181,566 clips (codec / telephone conditions)
- Phase 3 — ASVspoof 2021 DF: 611,829 clips (deepfake / neural TTS)
- Phase 4 — ASVspoof 2021 PA: up to 60k clips (physical access / replay)
- Fine-tune — Personal mic: 105 real + 38 high-confidence fake clips
- Total training exposure: > 640,000 labelled audio clips

Training Details

- Optimizer : AdamW | Loss: BCELoss | Mixed precision (AMP, GradScaler)
- LR schedule: cosine annealing with 2-epoch warmup
- Augmentation: gain, Gaussian noise (10–30 dB SNR), low-pass filter, codec sim
- SpecAugment: 2 freq masks (max 20 bins) + 2 time masks (max 30 frames)
- Class balancing: WeightedRandomSampler (counters 9:1 spoof/bonafide imbalance)
- Fine-tune: progressive unfreezing (FC-only phase → full network at 0.1× LR)

Live Pipeline

- Browser mic → 16 kHz float32 PCM → WebSocket (64 KB / 1-second frames)
- 3-second rolling buffer + 3-frame CNN score smoothing (deque maxlen=3)
- CNN + Whisper run concurrently via asyncio.gather every frame
- Fusion: score = 0.60×CNN + 0.30×keyword + 0.10×metadata
- Band thresholds: LOW ≤ 34 | MEDIUM 34-45 | HIGH > 45
- Actions: PASS | OTP_CHALLENGE | BLOCK

Active weights: voiceguard_cnn_finetuned.pt | Device: CUDA